

Data Analytics Essentials with AI: SQL, Python, Power BI, Tableau and Excel.

Project Report

Academic Task: 2 (Project)



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FIFA World Cup 2026 – Performance Analysis up to the Semi-finals

1. Introduction

Sports analytics has become an essential tool for understanding team and player performance. The FIFA World Cup 2026, the first tournament with 48 participating teams, generated a rich dataset that can be analyzed using Business Intelligence tools. This project uses Power BI to transform raw football statistics into an interactive dashboard that helps identify the factors contributing to success up to the semi-final stage.

2. Objectives

- Analyze the overall performance of teams in the tournament.
- Compare semi-finalists with other teams.
- Identify key attacking, defensive, and goalkeeping contributors.
- Measure team performance using custom performance indices.
- Present findings through an interactive four-page Power BI dashboard.

3. Problem Statement

Although detailed match and player statistics are available, it is difficult to identify the characteristics of successful teams without proper visualization. This project addresses that challenge by developing a dashboard that highlights the performance differences between semi-finalists and other teams.

4. Dataset Description

The project uses four datasets:

- matches.csv – Match-level statistics.
- teams.csv– Team performance statistics.
- players.csv– Individual player statistics.
- FactTeamMatch.csv– Cleaned fact table with one record per team per match.

The data includes possession, goals, shots, shots on target, saves, fouls, cards, corners, interceptions, and other performance metrics.

5. Methodology

The datasets were cleaned and transformed using Power Query. A star schema was created in Power BI, and DAX measures such as Team Attack Index, Team Defense Index, Match Control Index, and Team Performance Index were developed. Interactive visualizations and slicers were then used to explore team and player performance.

6. Dashboard Overview

The dashboard consists of four pages:

- Executive Overview: Tournament KPIs and top-performing teams.
- Team Performance Analysis: Comparison of semi-finalists with other teams using attack, defense, and possession metrics.
- Player Impact Analysis: Evaluation of top attacking, defensive, and goalkeeping performers.
- Executive Conclusion: Summary of the characteristics that defined successful teams.

7. Key Findings

- The analysis of the FIFA World Cup 2026 data up to the semi-final stage revealed several important patterns that distinguished the tournament's most successful teams. France, Argentina, and the Netherlands achieved the highest overall Team Performance Index, each recording a score of 0.75, indicating consistent excellence across attacking, defensive, and match control metrics.
- A comparison between semi-finalists and the remaining teams showed that the strongest performers consistently outperformed their opponents in Match Control Index, Average Possession, and Team Attack Index. This suggests that successful teams were able to dominate the flow of matches, create more attacking opportunities, and maintain greater control of possession throughout the tournament.
- The player-level analysis further highlighted the importance of individual quality in determining team success. Elite attacking players such as Kylian Mbappé, Lionel Messi, and Erling Haaland contributed significantly through goals, assists, and overall offensive influence, demonstrating that world-class attacking talent played a decisive role in advancing teams to the later stages of the competition.
- The findings also indicate that match control and possession were stronger indicators of success than traditional defensive statistics. While defensive organization remained important, teams that consistently controlled possession and dictated the tempo of matches were more likely to achieve positive results than those relying primarily on defensive actions.
- Interestingly, semi-finalists recorded lower numbers of saves and interceptions compared to many other teams. Rather than indicating weaker defensive performance, this reflects the fact that these teams conceded fewer attacking opportunities by controlling possession and limiting opposition attacks. As a result, their goalkeepers and defenders were required to perform fewer reactive defensive actions, reinforcing the conclusion that controlling the game was a defining characteristic of tournament success.

8. Conclusion

The analysis shows that successful teams combined strong attacking efficiency with superior control of possession and the overall flow of the game. Rather than relying solely on defensive actions, semi-finalists dominated matches through balanced team performance and effective finishing. The Power BI dashboard provides a clear and interactive way to understand these performance patterns, demonstrating the value of descriptive analytics in modern football.

9. Limitations

Analysis covers matches only up to the semi-finals.

No predictive or machine learning models were included.

Advanced tracking metrics such as Expected Goals (xG) and player movement data were unavailable.

10. Future Scope

Future work can extend the analysis to the complete tournament, compare multiple World Cups, integrate live match data, and include predictive analytics for performance forecasting.

FIFA WORLD CUP 2026

Performance Analysis up to the Semi-finals

4

Semi-finalists

2.92

Goals/Match

286

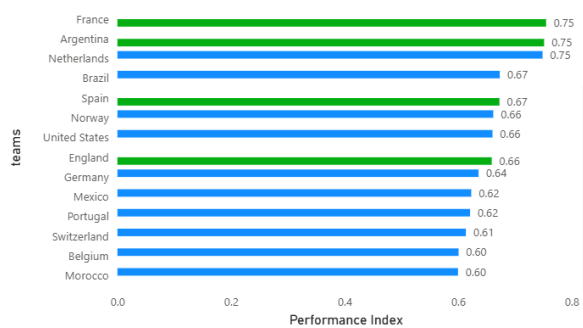
Goals Scored

0.54

Performance Index

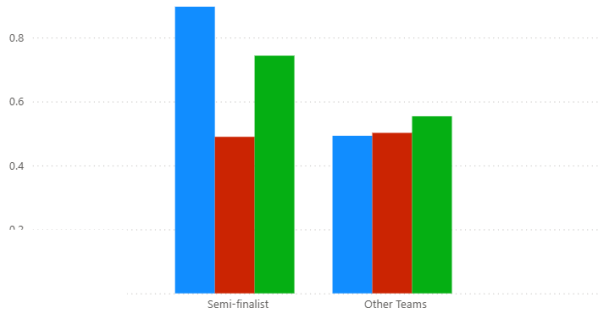
Top Performing Teams

Semi-finalist Other Teams Semi-finalist



Semi-finalists vs Other Teams

Match Control Index Team Defense Index Team Attack Index



KEY FINDINGS

- ✓ France and Argentina recorded the highest overall Performance Index (0.75).
- ✓ Semi-finalists outperformed other teams in Attack and Match Control.
- ✓ Defensive performance remained relatively consistent across both groups.
- ✓ Successful teams combined attack, defense and game control.

Team Performance Analysis

What performance characteristics separated the semi-finalists from the rest of the tournament?

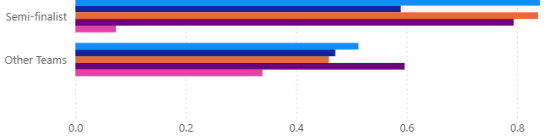
Semi-finalist Other Teams Semi-finalist

team
All

Round
☐ Group stage
☐ Round of 32
☐ Round of 16
☐ Quarter-finals

Result
All

Normalized Possession Normalized Goal Co... Normalized Goals ... Normalized Sh... Normalized S...



0.54

Performance Index

0.58

Team Attack Index

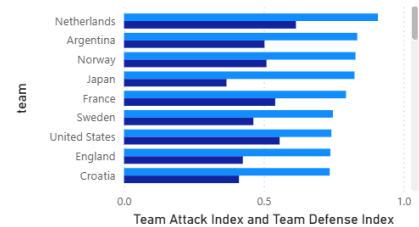
0.50

Team Defense Index

0.54

Match Control Index

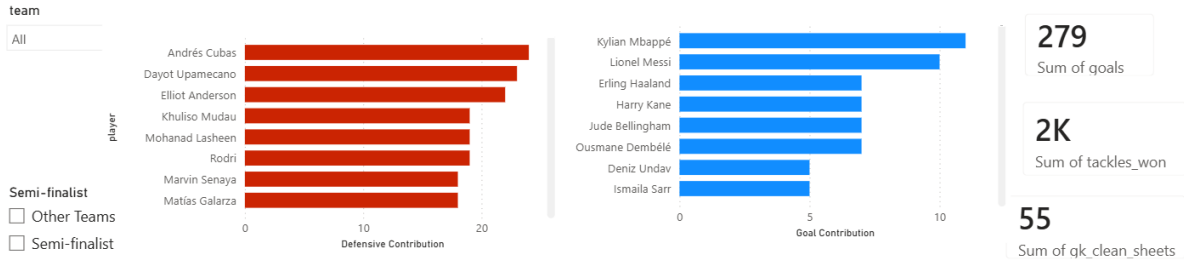
Team Attack Index Team Defense Index



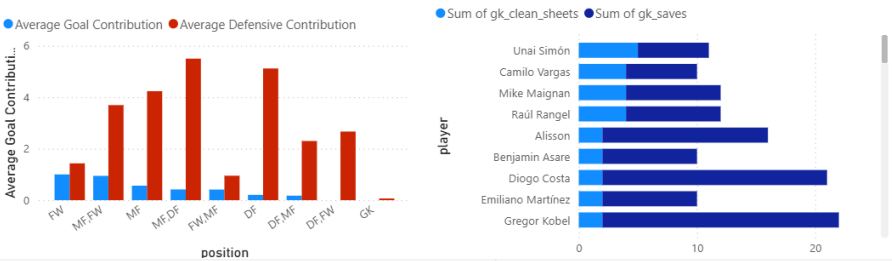
- Semi-finalists outperformed other teams across all normalized performance metrics, indicating consistent superiority rather than reliance on a single strength.
- Teams reaching the semi-finals combined above-average attacking quality with stable defensive performance, demonstrating the value of balanced gameplay.
- Attacking performance showed greater variation across teams than defensive performance, suggesting offensive efficiency was a stronger differentiator during the tournament.

Player Impact Analysis

Which players had the greatest influence on their team's performance across attacking, defensive, and goalkeeping roles?

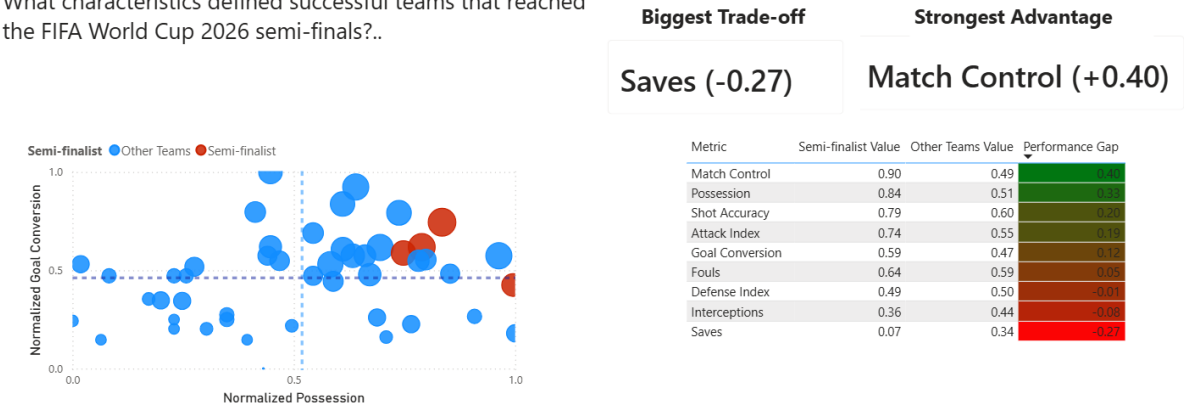


- 1. Attacking influence was concentrated among a small group of elite players.
- 2. Defensive contributions were distributed across more players than attacking contributions.
- 3. Goalkeeper performances varied considerably
- 4. Midfielders played the most balanced role.
- 5. Player impact extended beyond goals



What Defined Success? — The Semi-Finalist Profile

What characteristics defined successful teams that reached the FIFA World Cup 2026 semi-finals?..



- Key Insight,
- **Control drove the advantage** — Semi-finalists led by +0.40 in Match Control and +0.33 in Possession, giving them command of the game's tempo.
 - **Control converted into finishing** — That control translated into sharper attacking output: +0.20 in Shot Accuracy and +0.12 in Goal Conversion.
 - **Fewer defensive actions, not weaker defense** — Saves (-0.27) and Interceptions (-0.08) were lower for semi-finalists, but this reflects facing fewer shots and attacks overall, not a defensive gap.
 - **Takeaway** — Controlling the game, not just defending it, defined the path to the semi-finals.